

Temporal Drift

When Representational Time Separates from the Timing of Underlying Reality

A. Jacobs | Reality Drift Framework

Framework Context

Reality Drift describes the progressive loss of corrective contact between a system's representations and the real-world conditions they are meant to track, explain, or affect, even while the system remains functional and internally coherent. Temporal Drift is the time-based expression of that broader pattern.

Canonical Definition

Temporal Drift is the loss of alignment between the timing, pace, sequence, or memory of a representational system and the real-world processes it is meant to track. It appears when representations update on a different timescale than their referents, when records outlive the conditions that made them accurate, when feeds flatten sequence, or when new inputs overwrite the history needed to interpret the present.

What Is Drifting

Referent

The actual timing, duration, sequence, development, and consequence of real-world events or processes.

Operational representation

Feeds, dashboards, records, snapshots, timelines, forecasts, update cycles, and institutional memory.

Selection pressure

Systems reward recency, speed, refresh frequency, immediate legibility, and rapid response.

Weak correction

Slow outcomes, historical sequence, lagging consequences, and obsolete assumptions lack enough authority to alter present representations.

Representational Mechanism

Pace mismatch: A representation updates faster or slower than the condition it claims to track.

Sequence loss: Events are detached from the order and causal development required to interpret them.

Persistence mismatch: Records, categories, and models retain authority after the conditions that made them accurate have changed.

Memory overwrite: Successive updates displace the history needed to evaluate continuity, reversal, or contradiction.

Recursive Compression: Later summaries inherit temporally flattened accounts and reuse them as if their timing were still valid.

Feedback delay: Consequences arrive after the system has moved to a new cycle, weakening their ability to correct the decision that produced them.

Recognition Signs

- A dashboard updates frequently while the outcome it claims to represent develops slowly.
- Old records remain operationally authoritative after their source conditions have changed.
- Events recur in feeds without dates, sequence, or the context of earlier claims.
- Decisions are made from snapshots that conceal developmental or causal time.
- Consequences arrive too late to affect the cycle that produced them.
- The system remembers the latest state more clearly than the path by which it arrived there.

How Correction Weakens

Temporal correction requires more than current information. A system must preserve when a representation was accurate, what sequence produced it, how quickly its referent changes, and when consequences become observable.

Drift grows when update speed is mistaken for responsiveness. A rapidly refreshed representation can remain structurally stale if it measures the wrong temporal layer or erases the history needed to interpret change.

Cross-Domain Examples

Dashboards

Daily indicators are used to judge outcomes that develop over months or years, creating apparent responsiveness without temporal validity.

Policy

Categories and assumptions remain active after the conditions that justified them have changed.

Digital media

Old material resurfaces without its original date or sequence and enters the present as if it were current evidence.

AI systems

A model built from a frozen information environment is treated as current without reliable temporal qualification.

Institutional memory

The latest summary overwrites earlier uncertainty, reversals, and failed predictions, making the present account appear more continuous than the underlying history.

Boundary and Distinctions

Temporal Drift is not simply acceleration, distraction, asynchronous communication, or disagreement about timing. Different update rates can be appropriate.

Drift occurs when the representational timescale no longer matches the process being represented and that mismatch remains hidden or operationally uncorrectable. It is a structural desynchronization rather than merely a subjective feeling that life is moving quickly.

Relationship to Reality Drift

Temporal Drift is a time-based form of representational misalignment within Reality Drift. It weakens Semantic Fidelity by removing sequence, duration, and historical context.

It can support Narrative Drift when reordered events are fitted into a stable explanation, and Institutional Drift when records, categories, and models preserve an obsolete reality as the operative present.

Compact Diagnostic

1. What real-world process is this representation timing or sequencing?
2. Does the representation update on the same timescale as its referent?
3. What history is lost when the current state replaces the previous one?
4. When do consequences become visible, and can they still revise the originating decision?
5. Which records or categories have outlived the conditions that made them accurate?

The absence of failure is not proof of alignment. Coherence is not contact.